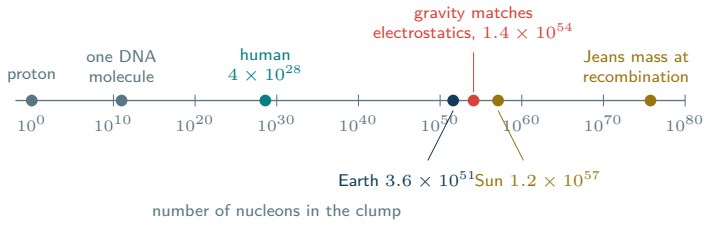
CLUMPING ON ONE PAGE

The making of Goldilocks: a story of clumping

Matter clumps when binding energy beats what pushes back, with Kleiber setting the terms for living bodies and Jeans for stars.

1 / THE RULE AND THE PICTURE

A merged whole weighs less than its parts, and the difference leaves as binding energy



Mergers of companies shed staff, economies of scale being economies of energy, and merging matter sheds binding energy in the same way. Three power laws then describe where a clump stops being passive:

$$W \propto M^{2/3}, \quad U \propto M^{3/4}, \quad M_J = \frac{p^{3/2}}{G^{3/2}\rho^2}$$

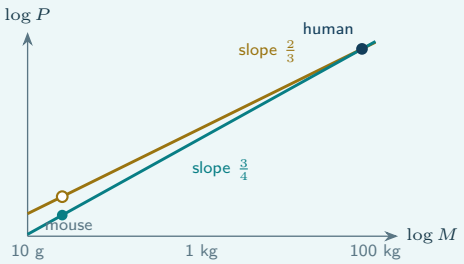
2 / INERT CLUMPS / WHEN SELF-GRAVITY MAKES A SPHERE

The work to remove one atom grows as $M^{2/3}$, so enough atoms overcome gravity's weakness

At constant density $M \propto r^3$, so $r \propto M^{1/3}$, and the work to lift one atom out of the body's own field is $W \propto M/r \propto M^{2/3}$. The electrostatic force between a proton pair exceeds gravity by $e^2/(4\pi\varepsilon_0 G m_p^2) = 1.24 \times 10^{36}$, so the gravitational gain $N^{2/3}$ catches up at $N \approx (10^{36})^{3/2} = 1.4 \times 10^{54}$ nucleons, which is 2.3×10^{27} kg, of Jupiter's order (the arithmetic $54 \times \frac{2}{3} = 36$). Sand grains and asteroids sit far below it and keep irregular shapes, while the Earth (3.6×10^{51} nucleons) and the Sun (1.2×10^{57}) bracket it.

3 / LIVING CLUMPS / THE INTERNAL CHEMICAL BUDGET OF KLEIBER

Heat made inside must leave through the surface, and the measured exponent $3/4$ fits a 3-sphere



A body that holds its temperature must metabolise at the rate it loses heat, so a 2-sphere with $A \propto r^2 \propto M^{2/3}$ predicts a $\frac{2}{3}$ power, whereas animals follow Kleiber's $U \propto M^{3/4}$. In d dimensions the boundary of a ball scales as $M^{(d-1)/d}$, so $\frac{3}{4}$ corresponds to $d = 4$, a 3-sphere, which suits the fractal-like internal structure of living bodies. Anchored at a 70 kg human using 82 W, the 25 g mouse draws 0.21 W by Kleiber against 0.41 W by the surface law, a human-to-mouse ratio of 385 where the area law gives 199.

4 / STELLAR CLUMPS / THE INTERNAL THERMAL PRESSURE OF JEANS

Gravity wins once GM^2/r exceeds the thermal energy pr^3, which fixes a critical mass

With $E_p \approx -GM^2/r$, internal energy $U \propto pr^3$ and $r = (M/\rho)^{1/3}$, clumping is favoured when $GM^2/r > pr^3$, giving $M_J = p^{3/2}/(G^{3/2}\rho^2) \propto \sqrt{p^3/\rho^4}$. At recombination, when the universe was 3000 K and nucleons and electrons first bound with the release of photons, $\rho = 9.9 \times 10^{-19}$ kg m^{-3} and the plasma pressure $p \approx \rho c^2/3 = 0.030$ Pa give

$$M_J = 9.6 \times 10^{48} \text{ kg} \approx 5 \times 10^{18} M_\odot$$

a ratio to the Sun of 4.8×10^{18}, comparable to the 5×10^{17} ratio of a 70 kg human to the mass of one of its chromosomes.

5 / TWO WAYS TO HOLD A CLUMP TOGETHER / THE PHOTON EATER AND THE KETOSIS

The human lives on photons from outside and the star lives on fuel stored inside

	human (Kleiber)	star (Jeans)
size	4×10^{28} nucleons	1.2×10^{57} nucleons
energy source	low-entropy photons through the food chain, a 500 nm photon in and thirty 15 μm photons out	hydrogen stored in the core and burnt from within, as in ketosis
what balances	heat produced against heat lost through the surface	gravitational pull against thermal and radiation pressure
scaling	$U \propto M^{3/4}$	$M_J \propto \sqrt{p^3/\rho^4}$

ASIDE / THE HUMAN IN THE MIDDLE

$\sqrt{m_p M_\odot} = 58$ kg, so a human is the geometric mean of a proton and the Sun, thriving between the quantum and the Newtonian.

ASIDE / CHECK IN PYTHON

`p**1.5/(G**1.5*rho**2)` with `p=rho*c**2/3` gives 9.56×10^{48} kg.

Convention. Counts are numbers of nucleons (M/m_p); scalings drop numerical factors. After *The making of Goldilocks: a story of Clumping* (McMurmerings, 2020).

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